DDR JEDEC Reading

DDR1 overview:

Architecture

Prefetch: 2n
Banks: 4
Voltage: 2.5V
Data transfer: rising + falling edge
Quad bank DRAM (4 banks)

Burst lengths: 2, 4, 8

Data signals

DQ -> Data Bus
DQS -> Data Strobe
DM -> Data Mask

Burst Based Access

DDR doesnt access single words. All accesses are burst oriented.

```
ACT → RD → burst data
ACT → WR → burst data
```

Address usage:

```
ACTIVE command
    BA0 BA1 → select bank
    A0-A13  → select row

READ/WRITE command
    A0-Ax → select starting column
```

Initialization

Before normal operation,

```
Power stable
Clock stable
wait 200 µs
```

Init sequence:

```
1. NOP / DES
2. PRECHARGE ALL
3. MRS (enable DLL via EMRS)
4. MRS (reset DLL)
5. wait 200 cycles
6. PRECHARGE ALL
7. 2 × AUTO REFRESH
8. MRS (final configuration)
```

Mode Register

Mode register defines operating parameters. Programmed using

```
MRS command
BA0 = 0
BA1 = 0
```

Fields:

```
A0–A2 → Burst Length
A3    → Burst Type
A4–A6 → CAS Latency
A7–A13 → Operating Mode
```

Commands

Commands are issued using these control signals.

```
CS#
RAS#
CAS#
WE#
```

Core commands

```
ACT       → open a row in a bank
READ      → read burst from active row
WRITE     → write burst to active row
PRE       → close row in bank
REFRESH   → refresh rows
MRS       → program mode register
NOP       → no operation
```

```
DES       → deselect device
BST       → terminate burst
```

Command flow

Typical operation sequence is:

```
ACT → RD → PRE
ACT → WR → PRE
```

Meaning:

```
1. Activate row
2. Access columns
3. Close row
```

Bank states

A bank can be in either Idle state or Row Active state.

Transitions occur this way:

```
Idle → ACT → Row Active
Row Active → PRE → Idle
```

Only when row is active you can issue:

```
READ
WRITE
```

ACTIVE Command

Purpose: Open a row in a bank

Address fields:

```
BA0 BA1 → bank selection
A0-A13  → row address
```

Rule: Only one row active per bank

To open another row: PRE → ACT

READ Command

Purpose: Start burst read from active row

Address fields:

```
BA0 BA1 → bank
A0-Ai   → column address
```

Important bit: A10 = Auto Precharge enable

Behavior:

```
A10 = 0 → row stays open
A10 = 1 → row closes after burst
```

WRITE Command

Same structure as READ.

Purpose: Start burst write

Data behavior:

```
DQ = data input
DM = data mask
```

DM logic:

```
DM = 0 → write data
DM = 1 → ignore data
```

PRECHARGE Command

Purpose: Close active row

Important bit:

```
A10 = 1 → precharge all banks
A10 = 0 → precharge selected bank
```

After PRE: bank becomes idle. Must wait for `tRP` before next ACT.

AUTO PRECHARGE

Feature used with:

```
READ
WRITE
```

Instead of issuing PRE manually:

```
 READ+ A10=1
WRITE + A10=1
```

Memory automatically performs `PRE` after burst

BURST TERMINATE

Used to: Stop ongoing READ burst early

Only applies when: Auto-precharge disabled

REFRESH

DRAM cells lose charge → must refresh.

Requirement: All rows refreshed every 64 ms

Example: 8192 rows → refresh every 7.8 μs

Timing parameter: `tREFI`

Command used: `AUTO REFRESH`

SELF REFRESH

Low-power refresh mode.

Used when: system sleep / standby

Behavior: internal refresh runs without external clock

Controller only toggles: `CKE`

Bank Command Rules (From Truth Tables)

Key idea: Commands depend on **bank state**.

When Bank = Idle

Allowed commands:

```
ACT
REFRESH
MRS
```

When Bank = Row Active

Allowed commands:

```
READ
WRITE
PRE
BST
```

Inter-bank Parallelism

Different banks operate **independently**.

Example:

```
ACT bank0
ACT bank1
READ bank0
WRITE bank1
```

This is how DRAM achieves **high throughput**.

Operations (Bank/Row Activation, Read, Write)

Row Activation

Before accessing data, a **row must be opened**.

ACT → opens row in selected bank

Address usage:

```
BA0, BA1 → bank
A0–A13   → row
```

After ACT, you must wait: `tRCD`

before issuing:

```
READ
WRITE
```

Reopening Rows

Only **one row can be active per bank**.

To open another row in the same bank:

`ACT → PRE → ACT`

Timing constraint:

`ACT → ACT (same bank) = tRC`

Bank Parallelism

Different banks operate independently.

Example:

```
ACT bank0
ACT bank1
```

Timing constraint:

`ACT → ACT (different bank) = tRRD`

This is how DRAM hides latency.

READ Operation

READ starts a **burst read** from an active row.

Address fields:

```
BA0 BA1 → bank
Column address → starting column
```

After issuing READ: wait CL cycles. Then data appears.

Data Transfer

Data is transferred using:

```
DQ  → data
DQS → data strobe
```

Transfer occurs on: both clock edges because of **DDR architecture**.

Burst Behavior

Example burst of 4:

```
READ
↓ CL cycles
D0 D1 D2 D3
```

Data appears on alternating edges of the clock.

Consecutive Reads

You can issue another READ while previous burst is running.

This allows: continuous data stream

Example:

`READ → READ → READ`

Burst Termination

A READ burst can be stopped early with: `BURST TERMINATE`

Used when: Auto-precharge disabled

READ → PRECHARGE

To close the row after reading: `READ → PRE`

Timing constraint: `READ → PRE = tRTP`

After PRE: wait `tRP` before next ACT.

WRITE Operation

WRITE starts a **burst write**.

Data behavior: data sent by controller

Data is captured on: edges of DQS

Write Data Mask

Signal: `DM`

Behavior:

```
DM = 0 → data written
DM = 1 → data ignored
```

Allows **partial writes**.

WRITE → READ

After a write, switching to read requires delay.

Timing parameter: `tWTR`

Meaning:

`WRITE → READ delay`

WRITE → PRECHARGE

To close the row after writing: `WRITE → PRE`

Timing constraint: `tWR`

After PRE: wait `tRP`

AC timing parameters (they define min delays between DRAM commands)

tRCD — Activate to Read/Write Delay

`ACT → RD / WR delay`

Meaning: After opening a row, the sense amplifiers need time before column access.

```
ACT
↓ tRCD
READ / WRITE
```

tRP — Precharge Delay

`PRE → ACT delay`

Meaning: Time required to close a row before opening another.

```
PRE
↓ tRP
ACT
```

tRAS — Active to Precharge Delay

ACT → PRE minimum time

Meaning: Row must stay active long enough to complete operations.

```
ACT
↓ tRAS
PRE
```

tRC — Row Cycle Time

`ACT → ACT (same bank)`

Relationship: tRC = tRAS + tRP

Meaning: Time between opening two rows in the same bank.

Bank Interaction Timing

tRRD — Row Activate to Row Activate (different banks)

`ACT bank A → ACT bank B`

Purpose: Limits how fast multiple banks can be activated.

Read / Write Switching

tWTR — Write to Read Delay

`WRITE → READ`

Needed when switching bus direction.

tWR — Write Recovery Time

`WRITE → PRE`

Time required to safely store written data before closing the row.

tRTP — Read to Precharge Delay

`READ → PRE`

Ensures read burst finishes before row closes.

Refresh Timing

tRFC — Refresh Cycle Time

REFRESH → next command delay

Memory unavailable during this period.

tREFI — Refresh Interval

Average interval between refresh commands

Example:

~7.8 μs for most DDR devices

DDR2 Overview:

DDR2 uses 4n prefetch architecture

Burst lengths supported - 4, 8

DDR2 Addressing Changes

Compared to DDR1:

```
Bank bits: BA0–BA2
Row address: A0–A15
```

Meaning DDR2 supports **more banks / larger address space**.

Command flow still remains: `ACT → READ / WRITE → PRE`

So the **core DRAM command model is unchanged from DDR1**.

DDR2 Initialization

Initialization is similar to DDR1 but with **more mode registers**.

Simplified sequence:

```
Power stable
Clock stable
NOP / DESELECT
PRECHARGE ALL
Program EMR registers
Reset DLL
PRECHARGE ALL
AUTO REFRESH (2+)
MRS (final configuration)
```

Important concept: All banks must be precharged before programming registers

DDR2 Mode Registers

DDR2 introduces **multiple mode registers**.

```
MR
EMR1
EMR2
EMR3
```

These control DDR2 features.

Mode Register (MR)

Controls core operating parameters.

Fields:

```
Burst Length
Burst Type
CAS Latency
DLL Reset
Write Recovery (WR)
Test Mode
```

New parameter added: WR (Write Recovery Time)

Extended Mode Register 1 (EMR1)

Introduces several **new DDR2 features**.

Controls:

```
DLL enable/disable
Output driver strength
Additive latency
ODT (On-die termination)
DQS disable
OCD control
RDQS enable
```

This register is **one of the biggest differences vs DDR1**.

DLL (Delay Locked Loop)

DLL must be enabled for normal operation.

Purpose: Align internal DRAM clock with external clock

Important rule: After DLL reset → wait 200 clock cycles before issuing READ. This allows clock synchronization.

Additive Latency (AL) — New DDR2 Feature

DDR2 introduces: Additive Latency (AL)

Purpose: Adds delay between ACT and READ to improve command scheduling

Effective read latency becomes: RL = AL + CL

Where:

RL = Read Latency
CL = CAS Latency

This allows better pipelining of commands.

On-Die Termination (ODT)

Another **major DDR2 addition**.

ODT = internal termination resistor

Purpose:

Reduce signal reflections
Improve signal integrity
Support higher frequencies

This becomes essential as DRAM speeds increase.

Extended Mode Registers

DDR2 introduces **multiple extended mode registers** to control advanced features.

```
EMR1
EMR2
EMR3
```

These registers configure **signal behavior, latency, and termination**.

EMR(2)

Purpose: Controls refresh-related features

Important rule: Must be programmed during initialization

Constraint: All banks must be precharged before EMR programming

Also must satisfy: `tMRD`

Mode Register Set delay.

EMR(3)

Reserved (no defined function)

Requirement: Must still be programmed during initialization. All reserved bits must be set to 0

OCD — Off-Chip Driver Calibration

DDR2 introduces **output driver calibration**.

Purpose: Adjust DRAM output driver impedance

Goal: Match transmission line impedance and Reduce signal reflections

Important characteristics: Default impedance ≈ 18 Ω

Calibration steps:

```
Drive mode
Adjust mode
Calibration exit
```

Controller must issue **EMRS commands** to perform calibration.

OCD Adjustment

Calibration procedure: Controller sends calibration code via DQ pins. DRAM adjusts output driver strength

Properties: Applies to all DQ and DQS signals, Up to 16 adjustment steps

Once calibration completes: Driver impedance stabilized

On-Die Termination (ODT)

Major DDR2 feature.

ODT = internal termination resistor inside DRAM

Purpose:

```
Reduce signal reflections
Improve signal integrity
Enable higher bus frequencies
```

Signals terminated:

```
DQ
DQS
DM
RDQS
```

Controller controls ODT via ODT pin

ODT Operation

ODT can be enabled during:

```
ACTIVE
STANDBY
```

ODT is disabled during: `SELF REFRESH`

ODT Update Timing

Changing ODT configuration requires delay `tMOD`

Rule: CKE must remain HIGH during tMOD. This ensures correct termination update.

Read / Write Access

After a bank is activated: `ACT → RD / WR`

Command encoding:

```
CS = LOW
CAS = LOW
RAS = HIGH
WE = HIGH → READ
WE = LOW  → WRITE
```

DDR2 still uses **burst-based access**. Each burst stays **within a fixed column segment of the page**.

Burst Boundaries

Example: Page size = 2048 bits

Segments:

BL = 4 → 512 segments
BL = 8 → 256 segments

Burst access must **remain within its segment**.

CAS-to-CAS Delay

Minimum delay between column commands:

`tCCD = 2 clocks`

Example:

```
READ → READ
```

```
WRITE → WRITE
```

Must wait **tCCD cycles**.

Posted CAS (Important DDR2 Feature)

DDR2 introduces **Posted CAS** to improve bus efficiency.

Problem in DDR1

In DDR1:

```
ACT
↓ tRCD
READ
```

Controller must wait **tRCD** before issuing READ.

DDR2 Solution

DDR2 allows:

ACT → READ immediately

The DRAM **internally delays the READ**.

This delay is called: Additive Latency (AL)

Additive Latency (AL)

Definition: Delay added between ACT and column command

Purpose: Allow command pipelining and Improve bus efficiency

The DRAM internally buffers the command.

Read Latency

In DDR2: RL = AL + CL

Where:

RL = Read Latency
CL = CAS Latency
AL = Additive Latency

Write Latency

Write latency is defined as: WL = RL - 1

So: WL = AL + CL - 1

Why Posted CAS Exists

Purpose: Allow ACT and READ commands to overlap, Improve command scheduling,Increase bandwidth. This is important for **high-frequency DDR2 operation**.

DDR3 Overview

DRAM Organization

DDR3 SDRAM is organized as an 8-bank DRAM.

Address mapping:

```
BA0–BA2 → bank
A0–A15 → row
Column address supplied during READ / WRITE
```

Command flow remains unchanged: ACT → READ / WRITE → PRE

So **core DRAM operation is the same as DDR1/DDR2**.

8n Prefetch Architecture

Major DDR3 change.

```
DDR1 → 2n prefetch
DDR2 → 4n prefetch
DDR3 → 8n prefetch
```

Meaning: 1 internal DRAM access → 8 data transfers externally

Data transfer behavior:

```
Internal core → 8n bits in 4 clock cycles
External I/O → 8 transfers on both clock edges
```

This enables **much higher bandwidth**.

Burst Operation

DDR3 accesses are still **burst oriented**.

Supported burst modes:

BL8 → burst length 8
BC4 → burst chop 4

Burst chop allows **shorter accesses without changing prefetch**.

Address Usage

During commands:

```
ACT command
    BA0–BA2 → select bank
    A0–A15 → select row
```

```
READ / WRITE command
    column address
    A10 → auto-precharge
    A12 → select BC4 or BL8
```

DDR3 State Machine

The state machine is almost identical to DDR2.

Main states:

```
Idle
Activating
Bank Active
Reading
Writing
Precharging
Refreshing
Power Down
Self Refresh
```

Transitions still follow: `ACT → RD/WR → PRE`

So **controller scheduling logic stays the same**.

DDR3 Initialization

Initialization sequence is similar to DDR2 but adds **ZQ calibration**.

Simplified sequence:

```
Power stable
RESET asserted
Release RESET
```

```
Clock stable
CKE HIGH
```

Then:

```
MRS → MR2
MRS → MR3
MRS → MR1
MRS → MR0 (DLL reset)
ZQCL (ZQ calibration)
Wait tDLLK + tZQinit
```

After this:

Memory ready for normal operation

ZQ Calibration (New DDR3 Feature)

DDR3 introduces **ZQ calibration**.

Purpose: Calibrate output driver impedance. Calibrate on-die termination resistance

Command used: `ZQCL`

Calibration uses an **external reference resistor (ZQ pin)**.

Mode Registers

DDR3 provides **four mode registers**:

```
MR0
MR1
MR2
MR3
```

They control:

```
burst length
CAS latency
additive latency
ODT
driver impedance
write leveling
DLL behavior
```

Mode registers are programmed using: MRS command

Requirements:

```
All banks must be precharged
tRP satisfied
CKE = HIGH
```

Minimum delay between MRS commands: tMRD

Mode Register MR0

MR0 controls **basic DRAM operation parameters**.

Fields include:

```
Burst Length
Burst Type
CAS Latency
DLL Reset
Write Recovery
Power-down behavior
```

Burst Length

Defined by: A0–A1

Supported modes:

```
BL8 (burst length 8)
BC4 (burst chop 4)
On-the-fly selection via A12
```

Burst order: Sequential and Interleaved

Selected via: A3

CAS Latency (CL)

Defined by: MR0 bits A9–A11

Definition: CL = delay between READ command and first data

DDR3 **does not support half-clock latencies**.

Read latency: RL = AL + CL

DLL Reset

Controlled by: MR0 bit A8

Behavior: DLL reset bit is self-clearing

Requirement: Must wait tDLLK after reset

Before issuing:

```
READ
ODT synchronous commands
```

Write Recovery (WR)

Defined in: MR0 bits A9–A11

Purpose: Minimum delay after WRITE before PRECHARGE

Relationship: tDAL = WR + tRP

Used during **auto-precharge after writes**.

Power-Down DLL Mode

Controlled by: MR0 bit A12

Two modes:

```
Slow exit  → DLL frozen
Fast exit  → DLL kept running
```

Exit timing:

```
Slow exit → tXPDLL
Fast exit → tXP
```

Mode Register MR1

MR1 configures **signal and interface behavior**.

Controls:

```
DLL enable/disable
Output driver impedance
RTT_NOM termination
Additive latency
Write leveling
TDQS
Output disable
```

Output Driver Impedance

Configured via: MR1 bits A1 and A5

Purpose: Match DRAM output impedance to transmission line and Improves signal integrity.

ODT (On-Die Termination)

DDR3 supports **two termination values**:

```
RTT_NOM
RTT_WR
```

Configured in:

```
RTT_NOM → MR1
RTT_WR → MR2
```

Purpose: Reduce reflections and Improve signal integrity

Additive Latency (AL)

DDR3 keeps the DDR2 concept.

Definition: AL = delay inserted between ACT and READ/WRITE

Read latency: RL = AL + CL

Write latency: WL = AL + CWL

Purpose:

```
Allow command pipelining
Improve bus efficiency
```

Write Leveling (Important DDR3 Feature)

DDR3 introduces **write leveling**.

Reason: DDR3 uses fly-by topology

This creates: clock arrival skew between DRAM chips

Write leveling allows the **controller to align DQS with the DRAM clock**.

Purpose: compensate clock-strobe skew and maintain correct write timing

Output Disable

Controlled by: MR1 bit A12

When enabled: DRAM output drivers disconnected

Used for: power measurement, debug

TDQS (Termination Data Strobe)

Available only in: x8 DDR3 devices

Purpose: provide additional termination resistance

Notes: shares pin with DM, not supported in x4/x16

Write Leveling

Why Write Leveling Exists

DDR3 introduces **fly-by topology** for command/address/clock routing.

Controller → DRAM1 → DRAM2 → DRAM3

This improves signal integrity but causes: clock arrival skew between DRAM chips

Meaning: DQS and CK arrive at different times

This can violate timing constraints:

```
tDQSS
tDSS
tDSH
```

So DDR3 introduces: Write Leveling

Purpose of Write Leveling

Goal: Align DQS with CK at the DRAM pins

This ensures correct **write timing**.

How Write Leveling Works

1️⃣ Controller enables leveling mode: `MR1[A7] = 1`

2️⃣ Controller sends **DQS pulses**.

3️⃣ DRAM samples: CK using rising edge of DQS

4️⃣ DRAM sends **feedback on DQ pins**.

5️⃣ Controller adjusts **DQS delay**.

6️⃣ When feedback changes: 0 → 1 transition

Controller locks the correct delay.

Result: DQS aligned with CK

Write Leveling Mode

Enabled via: MR1 bit A7 = 1

Exit leveling: MR1 bit A7 = 0

During leveling: Only NOP or DESELECT allowed

Special Behavior in Write Leveling

During leveling mode: Only DQS termination active

DQ termination remains: OFF

Controlled via: ODT pin

Byte Lane Leveling

For **x16 DRAM**:

```
Upper byte lane → separate feedback
Lower byte lane → separate feedback
```

So both lanes are leveled independently.

Self-Refresh Temperature Range (SRT)

DDR3 supports **temperature-dependent refresh behavior**.

Two operating modes:

```
Normal temperature
Extended temperature
```

Controlled via: `MR2[A7] → SRT bit`

Meaning:

```
SRT = 0 → Normal temperature refresh
SRT = 1 → Extended temperature refresh
```

Extended temperature requires: higher refresh rate

Auto Self-Refresh (ASR)

DDR3 can automatically adjust refresh behavior.

Enabled by: `MR2[A6] = 1`

When enabled:

```
DRAM automatically adjusts refresh rate
based on temperature
```

Benefits: lower power at low temperature and correct refresh at high temperature